2(a)	ρ: Nm / V	**B1**
	⅓: molecules move in three dimensions (not one) so ⅓ in any (one) direction	**B1**
	$<c^2>$: molecules have different speeds so take average	**M1**
	of (speed)^2	**A1**
2(b)	$pV = NkT$	**C1**
	$N = (3.0 \times 10^5 \times 6.0 \times 10^{-3}) / (1.38 \times 10^{-23} \times 290)$	**C1**
	$= 4.5 \times 10^{23}$	**C1**
	mass $= 20.7 / (4.5 \times 10^{23})$ $= 4.6 \times 10^{-23}$ g	**A1**

Question	Answer	Marks